LA MARTINIERE FOR BOYS, KOLKATA
HALF YEARLY EXAMINATION 2023-24

SUBJECT: COMPUTER SCIENCE **TIME:** 3 hours + 15 minutes reading time
CLASS: XI **FULL MARKS:** 70

Answer all questions in Part I (compulsory) and six questions from Part-II, choosing two questions from Section-A, two from Section-B and two from Section-C. All working, including rough work, should be done on the same sheet as the The intended marks for questions or parts of questions are given in brackets [ ]

Part 1

(Attempt all questions) **(20 marks)**

Question 1

a. Which of the following is a Universal Gate [1]

i. OR
ii. XOR
iii. AND
iv. NOR

b. If A and B are the inputs to any logic gate, then the output of which of the following logic gates will be AB+A'B' [1]

i. XNOR
ii. NAND
iii. XOR
iv. NOR

c. A proposition that has only false values (0) in its truth table column is a [1]

i. Tautology
ii. Converse
iii. Contradiction
iv. Inverse

d. The CONJUNCTION in case of propositional logic implies which of the following operations [1]

i. NOT
ii. OR
iii. AND
iv. IF....THEN

e. The keyword that is used to denote the object that is used to invoke a function [1]

i. this
ii. static
iii. final
iv. new

f. Define a well formed formula?
g. Find the complement of the Boolean expression AB + C'D' [1]
h. Find the dual of the following expression A.B + B'+1=1 [1]
i. State the contrapositive form of the expression A → B [1]
j. Write the canonical product of sum expression for the function f(A,B)=A.B [1]
[1]

Question 2:

a. Design a 2 input OR Gate using NAND gates. (Accept A and B as inputs) [2]
b. State De-Morgan's theorem for 2 variables A and B [2]
c. Consider the following code and answer the questions that follow. Show the dry run/ working

```
int check_num(int a)
  {
    int a1=0;
    int b=a;
    while(b!=0)
    {
      int c=b/10;
      int s=0;
      int d=b-c*10;
      a1=a1*10+d;
      b=b/10;
    }
    return a1;
          }
```

i) What will the function check_num return when 3243 is passed as an argument? (2)
ii) What is the function check_num doing? (1)

d. The following code is a method/function of a certain class. Give the output of the function solve when the value of ***n = 419***. Show the dry run / working. [3]

```
void solve( int n )
{
```

```
int t1=0, t2;
while( n>2)
{
if (n%2==1)
n*=5;
t2=n%7;
System.out.print("\n t2 = \t" + t2);
t1=t1+t2;
n/= 15;
}
System.out.print("\n t1 = \t" + t1);
}
```

PART – II

Answer six questions in this part, choosing two questions from Section A, two from Section B and two from Section C. **(50 Marks)**

Section A

Answer ***any two*** questions

Question 3:

Perform the following: [2 X 5 = 10]

a. $(110101.110)_2 = (?)_{10}$
b. $(AB19E)_{16} = (?)_8$
c. $(76523)_{10}=(?)_{16}$
d. $(110111)_2+(10111)_2$
e. $(11100101)_2 - (11011)_2$ using 2's complement form

Question 4:

a. A department intends to select employees for an on-site training as per the following criteria [5]

The employee is a graduate from any discipline and has been working in the company for more than 10 years

OR

The employee is either a graduate or an undergraduate but has won the best Employee award

P.T.O

The inputs are:

Inputs	
G	The person is a graduate
C	Has been working in the company for more than 10 years
E	Has won the best employee award

(In all the above cases, 1 indicates yes and 0, indicates No)

Output: X (1 indicates yes and 0 indicates no)

Draw the truth table for the inputs and outputs given above. Derive the terms with conjunctive operators for each of the true values (1's) from the output column. Also, write a propositional expression by joining terms with disjunction operators.

b. What is a half adder? Derive the expression of Sum and Carry using a truth table [3]

c. Simplify the following expression using Boolean laws [2]

M=XY+X(Y+Z)+Y(Y+Z)

Question 5:

a). Write one point of difference between a half adder and a half adder. Derive the expression of sum and carry of a full adder using truth table. [5]

b). Verify if the following proposition is a tautology or a contradiction [3]

[(a=>b)^(b=>c)]=>(a=>c)

c). State the principle of duality. [2]

SECTION – B

Answer any two questions.

Each program should be written in such a way that it clearly depicts the logic of the problem. This can be achieved by using mnemonic names and comments in the program. (Flowcharts and Algorithms are not required.) The programs must be written in Java.

Question 6: [10]

Design a class Unique to display the Unique words from a given String with the given member methods / member functions .

[A Unique word is a word which begins and ends with the same letter - eagle and legal are unique words]

Class name : Unique

Data members /instance variables

String n- to store the string

Member methods/ member functions

Unique() – initializes the data member n with default values
void input() – accept the String as an input from the user
boolean check(String b) – returns true if String 'b' is unique else returns false
void display() – displays all the unique words in a given string .

Write a function main() to create an object of the above class and call the member methods/ member functions

Question 7:

Design a class ColSum with the following member methods / member functions [10]

Class name : ColSum

Data members / instance variables

int a[][] – two dimensional array
int row – stores the number of rows
int col – stores the number of columns .

Member Methods / Member Functions

ColSum(int x, int y) – Initializes the data members row with the value of x , col with the value of y and the array with default values.
void input()-accept input in the given array.
int sum(int c) – returns the sum of each column 'c'.
void display() – displays the 2d array in a matrix form and the sum of each column by invoking the function sum .

Write a function main () to create an object of the above class and call the member methods/ member functions.

Specify the class ColSum giving details of the constructor(int,int), void input ( ),int sum(int) and void display( ). Define a main( ) function to create an object and call the functions accordingly to enable the task.

Question 8:

Design a class sumPrime to check whether the sum of the squares of the individual digits of a number is prime or not. [10]
Example: $12 = 1^2+2^2=1+4=5$ is a prime number
$81=8^2+1^2= 64 + 1=65$ is not a prime number.

Class name : sumPrime

Data members/Instance variables:

n- to store the number

sum- to store the sum of the square of the digits.

Member methods/functions:

sumPrime(int nn) – Initializes the number 'n' with nn and sum=0

void readdata()- accepts a positive integer.

int calc()- stores the sum of the square of the individual digits of the number

boolean prime(sum) – returns **true** if the sum is a prime number , else return **false.**

void display() – Displays an appropriate message by invoking the method boolean prime().

Specify the class sumPrime giving details of the constructor( ), void readdata( ),int calc(),boolean prime(int) and void display( ). Define a main( ) function to create an object and call the functions accordingly to enable the task.

SECTION – C

Answer any two questions.

Each program should be written in such a way that it clearly depicts the logic of the problem stepwise.

This can be achieved by using comments in the program

The programs must be written in Java and the algorithms must be written in general / standard form, wherever required / specified.

(Flowcharts are not required.)

Question 9: **[5]**

Design a class Fibonacci to display the nth fibonacci number.

Class name : Fibonacci

Data Members/Instance variables

int n- stores the nth number of the Series which is to be printed.

Member functions/Methods:

PrimeFibonacci(int x) – Parameterized constructor that initializes n with x

int generate(int x) – returns the xth Fibonacci number.

void display() – displays the xth fibonacci number by invoking the method generate(int)

Question 10:

A class MatPalin has been defined to display the palindrome numbers present in the two dimensional matrix **[5]**

Class name: MatAdd

Data members/Instance variables:

a[][]- to store integer elements
m: to store the number of rows
n: to store the number of columns

Member Functions/Methods:

MatPalin(int mm,int nn)-Parameterized Constructor to initialize the data members
void fill()- to enter the elements in the array
boolean Palin(int x) – returns true if x is a Palindrome number , else returns false .
void display() – displays all the Palindrome numbers in the array by invoking the function Palin(int)

The main function need not be written

Question 11:

i. Write a function int power(int x , int y) that computes and returns the value of x^y without the use of the library function Math.pow() **(3)**

ii. Define phishing **(2)**
